

AI Classroom Intelligence & Teaching Quality Platform

Engineering Product Blueprint (EPB) — v1.0

Document Class: Enterprise Product Blueprint (pre-IEEE-PRD stage) **Prepared for:** Engineering, AI/ML, Product, UI/UX, QA, DevOps, Security **Status:** Draft for Architecture Review **Scope Note:** This document is written sector-agnostic by design (K-12 / Higher Education / Corporate Training), with divergence points called out explicitly wherever legal, consent, or data-handling requirements differ by vertical. See §5.4 and §19 for the compliance matrix.

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1. Executive Summary

The AI Classroom Intelligence & Teaching Quality Platform ("the Platform") is an enterprise SaaS product that captures classroom lectures (audio, video, and presentation content), converts them into structured, analyzable data, and evaluates them along two independent axes:

1. **Curriculum Coverage** — did the lecture cover the right topics, in the right sequence, at the right depth, relative to the institution's approved syllabus.
2. **Technical Correctness** — were the concepts, definitions, formulas, and code presented actually accurate, verified against institution-approved reference material with citation-level evidence.

A third layer, **Teaching Quality Intelligence**, layers communication and instructional-design signals (clarity, example usage, structure) on top of the first two, and a fourth layer, **Faculty Improvement Center**, converts all of it into evidence-based, coaching-oriented feedback rather than a numeric score.

The Platform's defensible differentiation is Layer 2 (Technical Correctness). Competitors in this space (TeachFX, Sibme, Edthena, and similar classroom-analytics tools) measure *how* something was taught — pacing, talk ratio, questioning technique — but none verify *whether what was taught was true*. This is a genuinely open problem and the core engineering risk of this document (see §27).

The product is explicitly **not** built as a surveillance or ranking tool. This is a design constraint, not a marketing position — it shapes default visibility rules (§8), the Pilot Mode requirement (§7, Module 13), and the Dispute Workflow (§21), all of which are MVP-scope, not later additions.

2. Vision

To make continuous, objective, evidence-based teaching quality feedback as normal and non-threatening as a spell-checker — something that helps educators improve every lecture, rather than something that judges them after the fact.

The long-term vision is a platform that:

- Understands curriculum the way a subject-matter expert would.
 - Understands a lecture the way an attentive teaching assistant would.
 - Flags errors and gaps the way a rigorous peer reviewer would — always with evidence, never with a black-box verdict.
 - Converts all of the above into the kind of specific, actionable coaching a good department head would give, at a frequency no human department head could sustain (every lecture, not once a semester).
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3. Problem Statement

3.1 Background

Classroom teaching remains the primary medium of knowledge transfer, yet institutions have no reliable, objective, and scalable mechanism to evaluate how effectively — and how *accurately* — that teaching is delivered.

3.2 Current Challenges

Challenge	Description
Lack of continuous visibility	Classroom observations are episodic (once or twice a semester), not continuous.
Subjective evaluation	Student feedback and manual observation both carry significant rater bias.
No verification of technical accuracy	Existing tools assess presentation skill, not whether the content taught was correct.
Curriculum coverage is unverified	Institutions cannot systematically confirm topics were covered, in sequence, at appropriate depth.
Feedback is not actionable	Ratings and general comments, not specific, evidence-linked recommendations.
QA is manual and expensive	Academic quality assurance teams review lesson plans and reports by hand; does not scale.

3.3 Limitations of Existing Solutions

Existing classroom-analytics products (TeachFX, Sibme, Edthena, Otter-based meeting analytics) measure speaking pace, teacher/student talk ratio, question counts, and presentation confidence. None answer the central academic question:

"Was the subject matter taught correctly, completely, and according to the curriculum?"

3.4 Core Problem Statement

Educational institutions lack an AI-powered, evidence-based classroom intelligence platform capable of continuously determining whether a lecture aligns with the approved curriculum, whether the technical content is accurate, whether important topics are omitted or rushed, and what specific improvements would help faculty teach more effectively — without relying on fragmented, subjective, and manually intensive evaluation.

3.5 Why It Matters

Teaching quality directly affects student understanding, academic performance, graduate outcomes, and institutional accreditation. Small, continuous improvements compound over a semester in a way that infrequent, subjective feedback cannot produce.

4. Goals & Non-Goals

4.1 Goals

- G1: Provide continuous, automated curriculum-coverage tracking per lecture, course, and semester.
- G2: Provide evidence-based technical correctness validation with citation to source material and a confidence score on every flag.
- G3: Provide teaching-quality coaching signals that are explainable and tied to specific transcript timestamps.
- G4: Preserve and actively build faculty trust through consent, pilot mode, and dispute mechanisms as first-class, MVP-scope features — not post-launch additions.
- G5: Operate across at least one vertical (Higher Education, recommended MVP target — see §29) with an architecture that generalizes to K-12 and Corporate Training without a rebuild.

4.2 Non-Goals (explicitly out of scope)

- NG1: The Platform will **not** perform facial, posture, or emotion-based inference on students (attention estimation, confusion detection, hand-raise detection). Rejected on privacy, accuracy, and (for K-12) legal grounds — see §19.
- NG2: The Platform will **not** perform teacher behavioral surveillance (gesture analysis, eye-contact tracking, movement tracking). Undermines the core trust requirement (G4) for negligible signal value.
- NG3: The Platform will **not** produce a single numeric "teacher score" or public leaderboard/ranking in MVP or Phase 2. Any composite scoring is Phase 3 at the earliest, and only faculty-facing, never department-comparison-facing, without explicit institutional opt-in.

- NG4: The Platform will **not** perform predictive analytics (e.g., predicting which teachers will underperform) until sufficient historical data exists (Phase 3+, data-dependent, not date-dependent).
 - NG5: The Platform is not an LMS, attendance system, or HR/administrative-time tracker. Integrates with these systems; does not replace them.
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5. Target Users, Stakeholders & Personas

5.1 Target Users

- **Primary:** Faculty/teachers (direct users of feedback and self-view dashboard).
- **Secondary:** Academic administrators, department heads, quality-assurance teams.
- **Tertiary:** Institutional leadership / accreditation liaisons (aggregate, anonymized views only).

5.2 Stakeholders

Stakeholder	Interest
Faculty	Fair, accurate, non-punitive feedback that helps them improve
Department Heads	Visibility into curriculum coverage and content quality across sections
QA / Accreditation Teams	Auditable evidence of teaching standards for accreditation cycles
IT / Data Protection Officer	Data security, retention, and regulatory compliance
Institutional Leadership	ROI on faculty development, defensible accreditation evidence
Students (indirect)	Beneficiaries of improved teaching quality; not direct users or subjects of behavioral analysis

5.3 Personas

Persona	Role	Core Need	Core Fear
Dr. Meera Rao	Higher-ed Associate Professor, CS	Fast, specific feedback on whether her explanations are technically sound and well-paced	Being flagged unfairly by an AI with no way to contest it
Anand Verma	Head of Department, Engineering	Cross-section visibility into curriculum coverage consistency	Manually chasing faculty for lesson-plan compliance
Priya Nair	QA/Accreditation Officer	Auditable, evidence-linked records for accreditation review	Manual report compilation across departments
Institutional CIO	IT/Compliance	Data security, minor-safe handling if K-12 is in scope, vendor risk	A vendor breach or FERPA/DPDP-equivalent violation

5.4 Vertical Divergence Summary

Dimension	Higher-Ed	K-12	Corporate
Subject consent complexity	Faculty consent only	Faculty + parental/guardian consent for any student audio capture	Employee consent (lower friction)
Regulatory sensitivity	Moderate	High (minors — treat as highest-sensitivity default)	Low
Recommended MVP target	Yes	No — Phase 2+ after legal review	Possible parallel-track, lower strategic priority
Student-facing audio analytics	Permitted with consent	Recommend disabling by default	Permitted

6. Product Scope & Phased Feature Map

#	Feature	Module	Phase
1	Syllabus upload & parsing	Curriculum Intelligence	MVP
2	Reference material upload & indexing	Curriculum Intelligence	MVP
3	Live audio recording	Classroom Capture	MVP
4	Live video recording	Classroom Capture	MVP
5	Smart board / slide capture	Classroom Capture	MVP
6	Speech-to-text	Speech Intelligence	MVP
7	Speaker diarization	Speech Intelligence	MVP
8	Timestamped transcript generation	Speech Intelligence	MVP
9	Topic segmentation	Lecture Intelligence	MVP
10	Concept/definition/formula/code extraction	Lecture Intelligence	MVP
11	Lecture summary generation	Lecture Intelligence	MVP
12	Curriculum coverage tracking (covered/missed/partial/rushed)	Curriculum Intelligence	MVP
13	Concept/definition/formula/code correctness validation	Technical Validation	MVP
14	Reference-material comparison with citation	Technical Validation	MVP
15	Evidence-based error flagging + confidence score	Technical Validation	MVP
16	Human review queue	Technical Validation	MVP
17	Explainable-output layer (evidence+timestamp+confidence on every flag)	Cross-cutting principle	MVP

#	Feature	Module	Phase
18	Consent & opt-in management	Privacy & Governance	MVP
19	Faculty self-view dashboard	Reporting	MVP
20	Basic PDF lecture report	Reporting	MVP
21	Role-based access control	Privacy & Governance	MVP
22	Pilot / sandbox mode (scores hidden from admins)	Privacy & Governance	MVP
23	Topic sequence detection	Lecture Intelligence	Phase 2
24	Lesson-plan upload & learning-outcomes extraction	Curriculum Intelligence	Phase 2
25	Course timeline mapping	Curriculum Intelligence	Phase 2
26	Multi-language speech support	Speech Intelligence	Phase 2
27	Communication quality (pace, clarity, filler words)	Teaching Quality	Phase 2
28	Classroom interaction analytics (talk ratio, wait time, questions)	Interaction Intelligence	Phase 2
29	Time utilization analytics	Time Intelligence	Phase 2
30	Dispute / appeal workflow	Governance	Phase 2
31	Bias/fairness audit on correctness engine	Governance	Phase 2
32	Encryption, audit logs, retention policy automation	Privacy & Governance	Phase 2

#	Feature	Module	Phase
33	Faculty Improvement Center (trends, strengths/weaknesses)	Coaching	Phase 3
34	Weekly/monthly/semester reports	Reporting	Phase 3
35	Department & institution dashboards	Admin	Phase 3
36	Student learning insights (predicted difficulty topics)	Coaching	Phase 3
37	Assessment Intelligence (auto quiz/viva/Bloom's questions)	Coaching	Phase 3

Explicitly cut (see prior analysis, retained here for traceability): multi-camera support, student attention/confusion/hand-raise detection, teacher movement/eye-contact/gesture analysis, classroom noise analysis, administrative time/break analysis, predictive analytics (pre-Phase-3), voice-based "confidence" detection, standalone keyword identification.

7. Detailed Feature Specifications

Each MVP feature below follows the full specification template. Phase 2/3 features use a condensed template (full detail deferred to their respective design-review cycle).

7.1 Module: Curriculum Intelligence

7.1.1 Syllabus Upload & Parsing (MVP)

Field	Detail
Purpose	Convert an institution's official syllabus into a structured, machine-readable curriculum tree.
Why it exists	Every downstream comparison (coverage, correctness) requires a ground-truth curriculum structure; without it the system has nothing to validate against.
User benefit	Admins upload once per course; no manual curriculum re-entry.
Inputs	PDF/DOCX syllabus, lesson plan (optional)
Outputs	Structured JSON: units → chapters → topics → learning outcomes → expected teaching hours
AI models	LLM (GPT-4-class or equivalent) for semantic structuring; OCR fallback for scanned documents
Tech stack	PDFPlumber/PyMuPDF, python-docx, Tesseract/Document AI, LLM API
APIs/services	Internal Curriculum Service; LLM provider API
Data stored	Structured curriculum JSON, source document (encrypted at rest)
Edge cases	Non-standard syllabus formats; scanned/handwritten syllabi; multi-course combined documents
Risks	Incorrect structuring silently propagates errors into every downstream module — requires admin confirmation step before activation
Future enhancements	Auto-detect curriculum standard (e.g., ABET, NBA) and map directly to accreditation criteria

7.1.2 Reference Material Upload & Indexing (MVP)

Field	Detail
Purpose	Ingest textbooks, faculty notes, and approved slide decks as the ground-truth corpus for correctness validation.
Why it exists	Technical Validation (§7.3) cannot function without an authoritative source to compare against — this is non-negotiable infrastructure.
User benefit	Institutions control exactly what "correct" means for their courses.
Inputs	PDF/DOCX/PPTX reference documents
Outputs	Chunked, embedded vector index per course, with page/edition metadata preserved for citation
AI models	Embedding model (e.g., text-embedding-3-large or equivalent); chunking heuristics
Tech stack	LangChain/LlamaIndex, ChromaDB or Pinecone, embedding API
Data stored	Vectorized chunks + original documents + page/edition metadata
Edge cases	Multiple conflicting editions of the same textbook; institution-specific notation deviating from the textbook
Risks	Poor chunking degrades citation precision, which degrades the credibility of every correctness flag
Future enhancements	Automatic edition-conflict detection and admin resolution UI

7.2 Module: Classroom Capture

7.2.1 Live Audio Recording (MVP)

Purpose	Capture lecture audio for downstream speech processing.
Why	Primary data source for the entire pipeline.
Inputs	Classroom microphone stream
Outputs	Timestamped audio file (encrypted)
Tech stack	MediaRecorder API / WebRTC, FFmpeg for transcoding
Edge cases	Multiple simultaneous speakers, background noise, microphone failure mid-lecture
Risks	Recording without active, verified consent is a hard blocker — see §7.4.1, must gate recording start

7.2.2 Live Video Recording (MVP)

Purpose	Capture classroom video for slide-sync and teacher-side context (not student behavioral analysis — see NG1/NG2).
Inputs	Classroom camera (fixed or RTSP)
Outputs	Timestamped video file
Tech stack	OpenCV, RTSP ingestion, FFmpeg
Edge cases	Camera occlusion, poor lighting, no camera hardware installed

7.2.3 Smart Board / Slide Capture (MVP)

Purpose	Capture what's on-screen to compare against what's being said (slide-speech sync).
Inputs	Screen-share stream or periodic screenshot capture, uploaded PPTX
Outputs	OCR'd slide text with timestamps
Tech stack	python-pptx, OCR, screen-capture API
Note	Subsumes "board usage / PPT usage analysis" from earlier drafts — same underlying signal, one pipeline.

7.3 Module: Speech Intelligence

7.3.1 Speech-to-Text (MVP)

Purpose	Convert lecture audio into text.
AI models	Whisper (large-v3 or equivalent), with domain-adapted vocabulary injection for technical terms
Risks	Domain jargon, code spoken aloud, and accented technical English are the single largest source of downstream error — errors here corrupt every later stage, including correctness validation. This is treated as a PO accuracy risk, not a Phase 2 polish item.

7.3.2 Speaker Diarization (MVP)

Purpose	Separate teacher speech from student speech.
Tech stack	pyannote.audio / WhisperX
Edge cases	Overlapping speech, students far from mic, guest lecturers

7.3.3 Timestamped Transcript Generation (MVP)

Purpose	Produce the canonical, searchable transcript that every other module references by timestamp.
Output	JSON transcript: {speaker, text, start_ts, end_ts, confidence}

7.4 Module: Lecture Intelligence

7.4.1 Topic Segmentation (MVP)

Purpose	Divide the transcript into topic-coherent segments.
AI models	Sentence-transformer embeddings + LLM-based boundary detection
Output	List of {topic_label, start_ts, end_ts}

7.4.2 Concept/Definition/Formula/Code Extraction (MVP)

Purpose	Pull out every discrete claim the correctness engine will need to check.
AI models	LLM extraction (structured output), spaCy for entity/term tagging
Output	List of atomic claims, each linked to a transcript timestamp
Note	This is the direct input to Technical Validation (§7.5) — extraction quality caps validation quality.

7.4.3 Lecture Summary (MVP)

Purpose	Generate a concise summary + key takeaways for the faculty self-view dashboard.
AI models	LLM summarization
Note	Merged with "key takeaways" — same output, not two features.

7.5 Module: Curriculum Coverage Intelligence (MVP)

Purpose	Compare segmented lecture topics against the structured curriculum to compute coverage status.
Outputs	Per-topic status: Covered / Missed / Partially Covered / Rushed; time-per-topic; overall coverage %.
AI models	Semantic similarity matching (embeddings) between lecture topics and curriculum topics
Tech stack	Curriculum matching engine, vector similarity search
Edge cases	Curriculum topic taught out of order but fully covered; topic covered across multiple lectures

7.6 Module: Technical Content Validation (Core USP) (MVP)

Field	Detail
Purpose	Determine whether extracted claims (concepts, definitions, formulas, code) are factually and pedagogically correct relative to approved reference material.
Why it exists	This is the product's defensible differentiator — no competitor verifies content correctness, only delivery style.
Inputs	Atomic claims from §7.4.2, vectorized reference corpus from §7.1.2
Outputs	Per-claim verdict: Correct / Incorrect / Ambiguous, with citation (source, edition, page), confidence score (0-1), and reasoning trace
AI models	RAG pipeline (retrieval + LLM judgment), citation-grounded generation
Tech stack	LangChain, ChromaDB, LLM (GPT-5-class/Gemini-class), custom citation engine
Edge cases	Valid alternate teaching methods flagged as "incorrect"; intentional pedagogical simplification; ambiguous or context-dependent correctness
Risks	False positives are the primary product risk. Every "Incorrect" verdict below a confidence threshold (configurable, default ≥0.85) must route to Human Review (§7.6 below), never surface directly to faculty as a claim of error.
Mitigation	Confidence scoring, mandatory citation, human review queue, and dispute workflow are bundled as one inseparable MVP unit — none may ship without the others.

7.6.1 Human Review Queue (MVP)

Purpose	Route low-confidence or high-impact flags to a human reviewer (subject-matter expert or department head) before they reach the faculty member as a finding.
Output	Reviewed verdict: Confirmed / Overturned / Needs More Context

7.7 Module: Privacy & Governance (MVP — elevated from later phase per architecture decision in §19)

7.7.1 Consent & Opt-In Management (MVP)

Purpose	Gate all recording behind explicit, revocable consent.
Why	Legally necessary in every vertical; the single largest trust lever for faculty adoption.
Behavior	Recording cannot start until faculty (and, where applicable, institutional parental-consent process for K-12) has actively opted in for that session; opt-out is always available and logged.

7.7.2 Pilot / Sandbox Mode (MVP)

| Purpose | New institutions run the Platform for a full term with all scores/flags visible only to the individual faculty member — never to administrators — before the institution can enable admin visibility. | | Why | Converts "this is surveillance" into "this is a tool I control," structurally, not just through messaging. |

7.7.3 Role-Based Access Control (MVP)

| Purpose | Enforce the visibility matrix defined in §8. | | Tech stack | OAuth 2.0 / JWT, PostgreSQL row-level security |

7.8 Module: Reporting (MVP)

- **Faculty Self-View Dashboard** — coverage %, flagged items with evidence, summary, trend over own lectures only.
- **Basic PDF Lecture Report** — exportable single-lecture report for the faculty member's own records.

7.9 Phase 2 Features (condensed specification)

Feature	Purpose	Key Tech	Key Risk
Topic sequence detection	Flag disorganized/out-of-order teaching	Segmentation + curriculum order graph	Low — cheap add over existing segmentation
Lesson-plan upload & LO extraction	Compare intended vs. actual coverage	LLM extraction	Format variance
Course timeline mapping	Visualize planned vs. actual pacing across semester	Timeline engine	Low
Multi-language speech support	Serve non-English-medium institutions	Whisper multilingual, translation layer	Correctness validation harder cross-language — must not silently degrade confidence scores
Communication quality	Pace, clarity, filler-word coaching	Whisper + NLP	Commodity capability — do not over-invest engineering time
Classroom interaction analytics	Talk ratio, wait time, question counts	Diarization + NLP	Proven model (TeachFX precedent) — low risk
Time utilization analytics	Teaching/discussion/silent time breakdown	Timestamp analytics	Low
Dispute/appeal workflow	Faculty formally contests a flag, routes to human review	Workflow engine	Must ship before any admin-visible scoring goes live institution-wide
Bias/fairness audit	Detect systematic bias by accent/language/style in the correctness engine	Statistical audit tooling on flag distribution	Requires accumulated flag data; needs its own eval framework
Encryption/audit logs/retention automation	Compliance infrastructure at scale	AES-256, PostgreSQL audit tables	Must precede any multi-institution rollout

7.10 Phase 3 Features (condensed specification)

Feature	Purpose	Key Risk
Faculty Improvement Center	Longitudinal strengths/weaknesses, progress tracking	Needs months of data to be meaningful — do not build empty-shell version early
Weekly/monthly/semester reports	Aggregate reporting	Low
Department & institution dashboards	Admin-level aggregate visibility	Must remain anonymized/aggregate per pilot-mode graduation rules
Student learning insights	Predicted difficulty topics, likely misunderstandings	Derived from correctness + coverage engines — must mature first
Assessment Intelligence	Auto-generate quizzes/viva/Bloom's-taxonomy questions	Genuine differentiator vs. TeachFX/Sibme — reframes product as teaching tool, not audit tool